

orjinal

$F_s = 200\ Hz$

tamsayı

$T_{yeni} = MT = T'$

$F_{s_{yeni}} = 100\ Hz \quad M = 2$

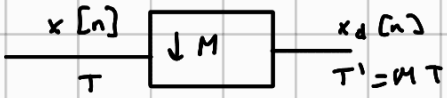


$x[n] = x(nT)$

$x_d[n] = x[nM]$

down samples

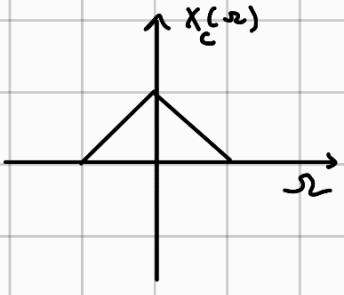
	$M=2$	$M=3$
$x[0] = A$	$x_d[0] = A$	$x_d[0] = A$
$x[1] = B$	$x_d[1] = C$	$x_d[1] = D$
$x[2] = C$	$x_d[2] = E$	$x_d[2] = G$
$x[3] = D$		
$x[4] = E$		
$x[5] = F$		
$x[6] = G$		

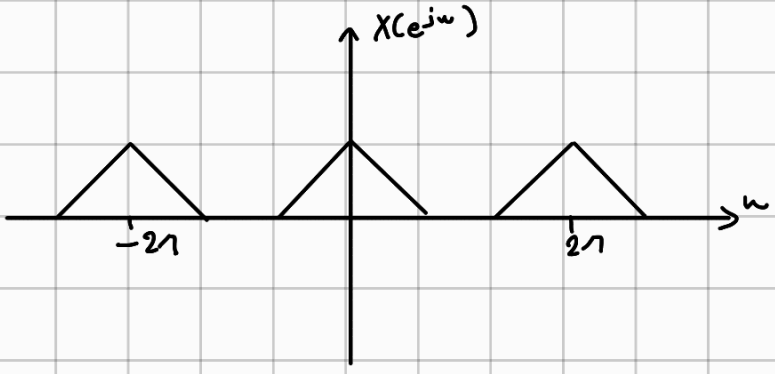


$$x[n] \longleftrightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

($\frac{\omega}{T} = \Omega$, $\frac{2\pi k}{T} = \Omega_k$)





$$X_d(e^{j\omega}) = \frac{1}{T'} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T'} - \frac{2\pi k}{T'}\right)\right)$$

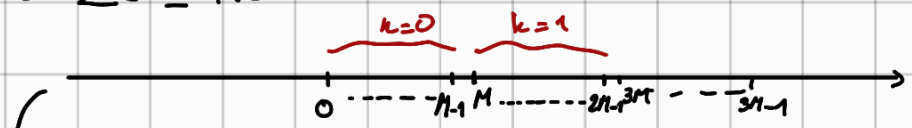
$$= \frac{1}{MT} \sum_{r=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{MT} - \frac{2\pi}{MT} r\right)\right)$$

r yazalım

$r = i + kM$

$0 \leq i \leq M-1$

$-\infty < k \leq +\infty$



→ Görüldüğü üzere $r = -\infty$ ile $+\infty$ arasında değer alıyor

$$X_d(e^{j\omega}) = \frac{1}{MT} \sum_{r=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{MT} - \frac{2\pi r}{MT}\right)\right)$$

$$= \frac{1}{M} \sum_{i=0}^{M-1} \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{MT} - \frac{2\pi(i+kM)}{MT}\right)\right)$$

$$\frac{2\pi i + 2\pi kM}{MT} = \frac{2\pi i}{MT} + \frac{2\pi k}{MT}$$

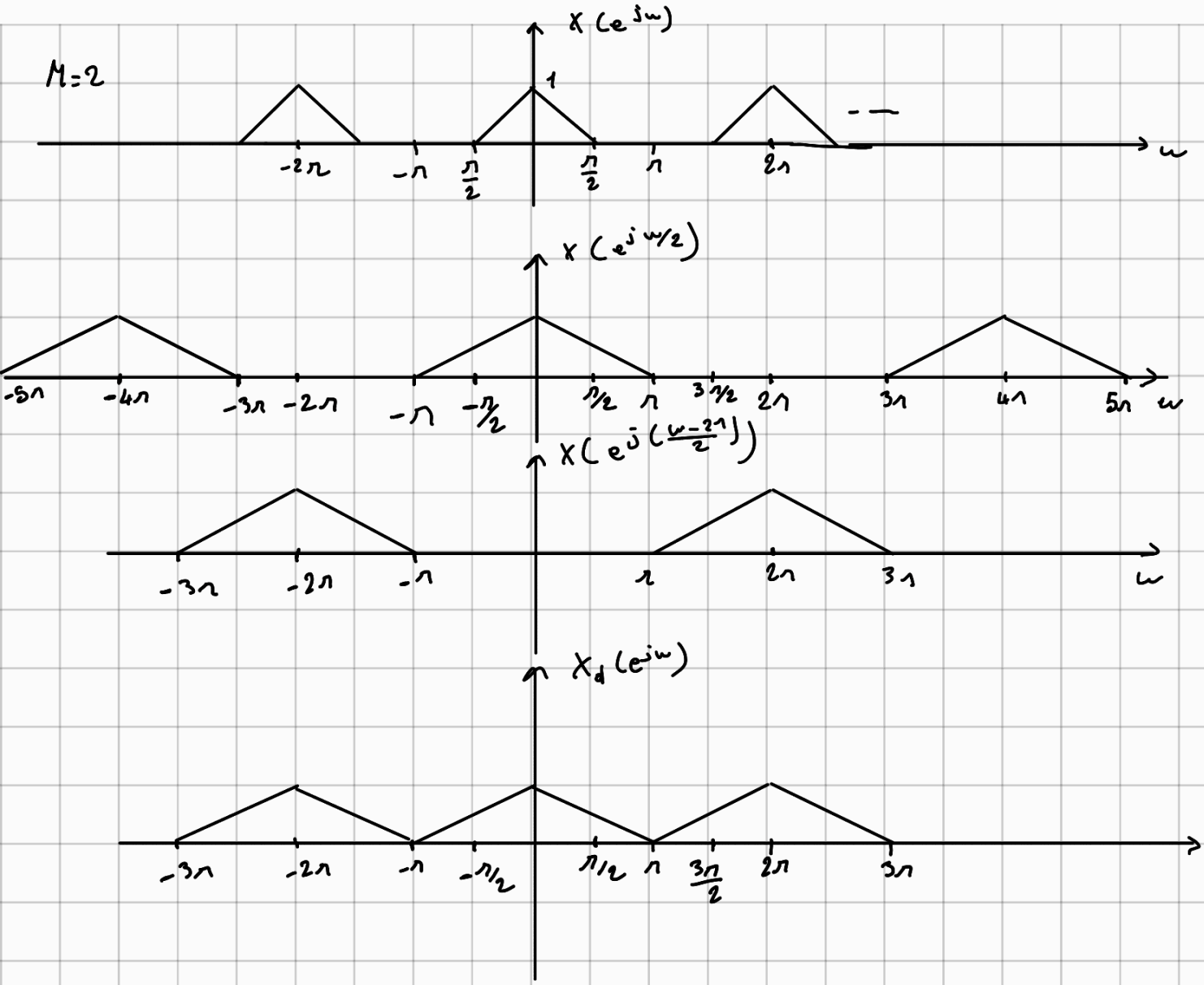
$$= \frac{1}{M} \sum_{i=0}^{M-1} \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi i}{MT} - \frac{2\pi k}{T}\right)\right)$$

$$X_d(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} X(e^{j\omega})\Big|_{\omega = \frac{\omega - 2\pi i}{M}}$$

$$X_d(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} X\left(e^{j\left(\frac{\omega - 2\pi i}{M}\right)}\right)$$

$M=2 \rightarrow X_d(e^{j\omega}) = \frac{1}{2} X\left(e^{j\frac{\omega}{2}}\right) + \frac{1}{2} X\left(e^{j\left(\frac{\omega - 2\pi}{2}\right)}\right)$

$X_d(e^{j\omega}) = \frac{1}{3} X\left(e^{j\frac{\omega}{3}}\right) + \frac{1}{3} X\left(e^{j\frac{\omega - 2\pi}{3}}\right) + \frac{1}{3} X\left(e^{j\frac{\omega - 4\pi}{3}}\right)$



$M = 2$ için $BG = \frac{\pi}{2}$

$M = 3$ " $\frac{\pi}{3}$

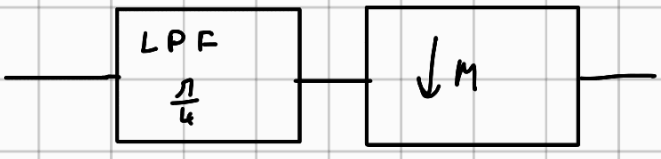
M " $BG = \frac{\pi}{M}$ olmalı,



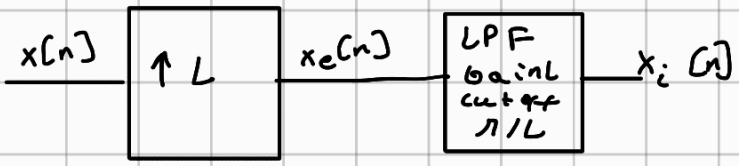
frekans artır,

* İşaretin bant genişliği en fazla $\frac{\pi}{M}$ kadar olmalı,
Seyrek örneklediğinde geçişler keskinleşiyor frekans artıyor.

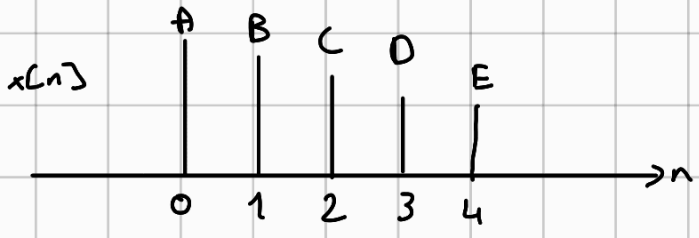
Down sampling



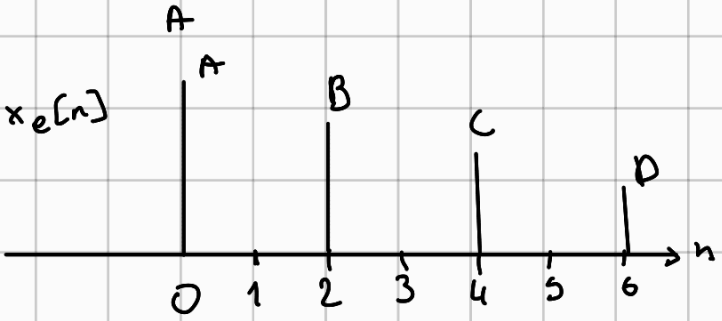
Up sampling



$$x_e[n] \triangleq \begin{cases} x\left[\frac{n}{L}\right], & n = 0, \pm L, \pm 2L, \pm 3L \\ 0, & o.w \end{cases}$$

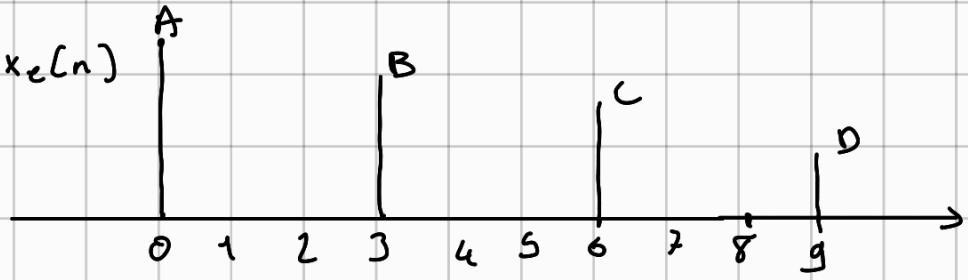


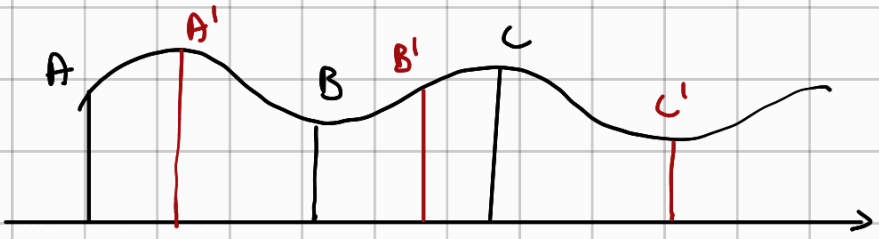
L = 2



araya L-1 ekliyor

L = 3





$$x_e[n] = \sum_{k=-\infty}^{\infty} x[k]\, \delta[n-kL]$$

$$X_e(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_e[n]\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k]\, \delta[n-kL] \right) e^{-j\omega n}$$

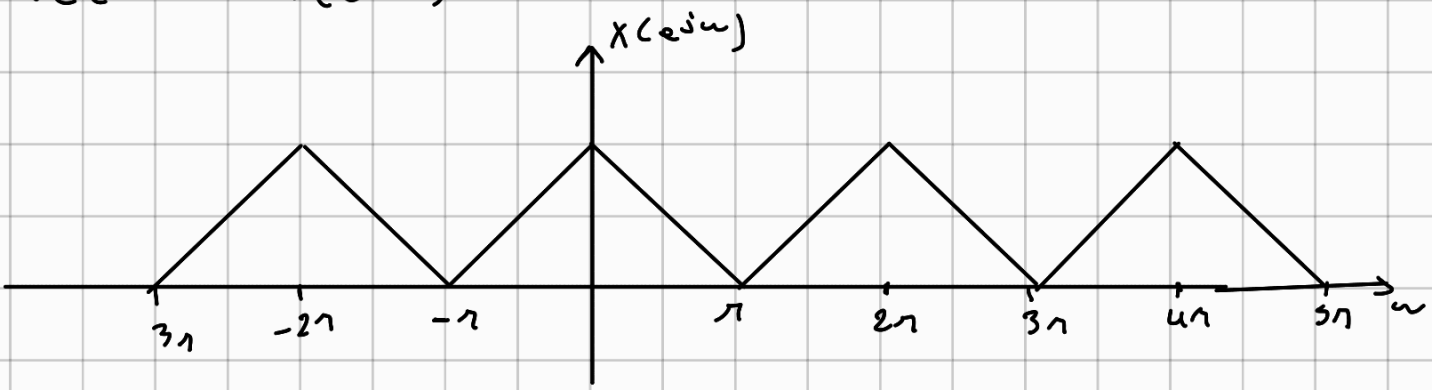
$$= \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} \delta[n-kL]\, e^{-j\omega n}$$

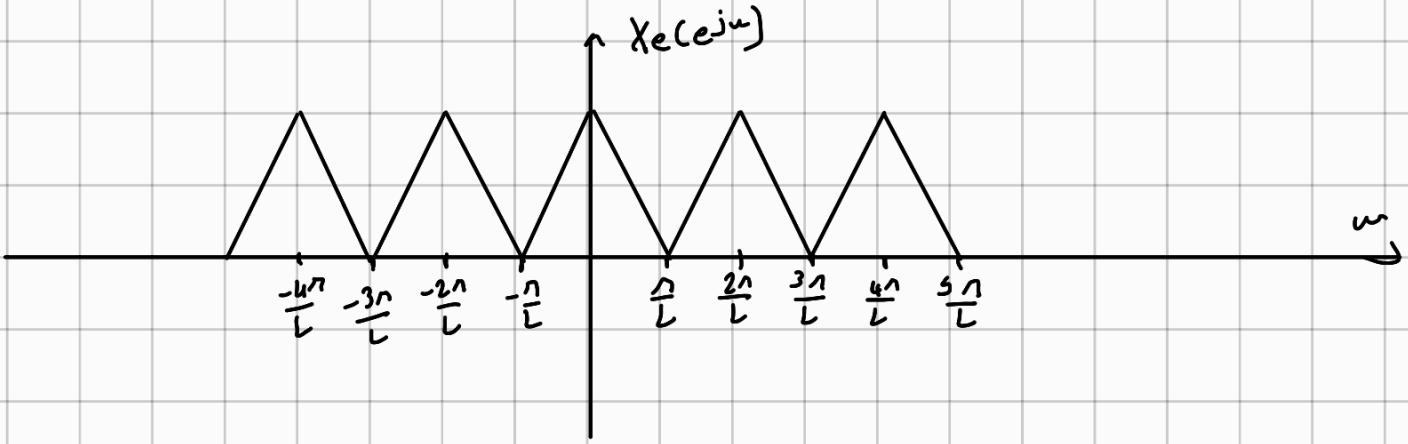
$$X_e(e^{j\omega}) = \sum_{k=-\infty}^{\infty} x[k]\, e^{-j\omega Lk}$$

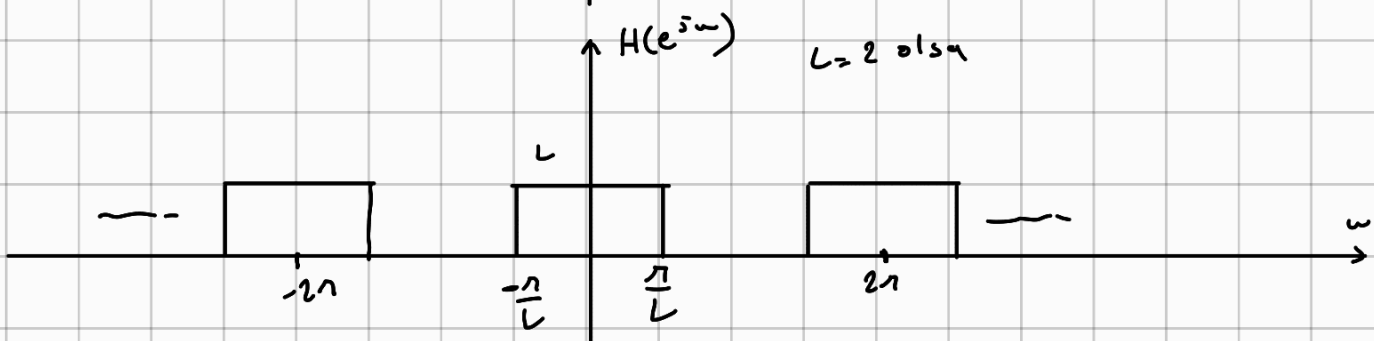
$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} x[k]\, e^{-j\omega k}$$

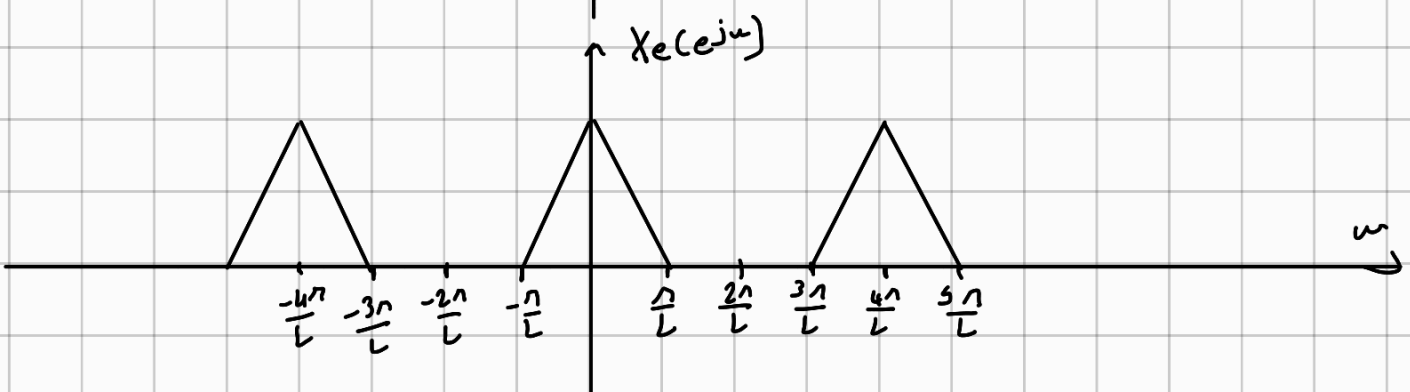
$$X_e(e^{j\omega}) = X(e^{j\omega}) \Big|_{\omega \leftarrow \omega L}$$

$$X_e(e^{j\omega}) = X(e^{j\omega L})$$









$$x_i[n] = x_e[n] * h[n]$$

$$h[n] = \frac{\sin \pi n/L}{\pi n/L}$$

$$x_i[n] = \sum_{k=-\infty}^{\infty} X_e[k]\, h[n-k] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-kL]$$

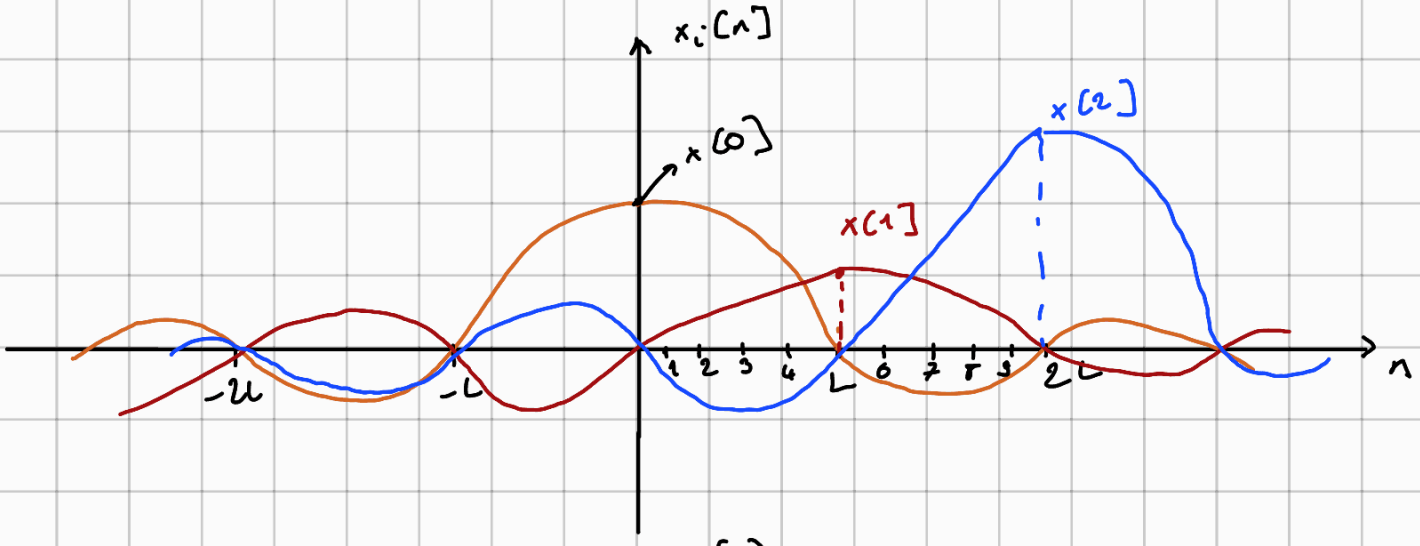
$$= \cdots + \underbrace{X_e[0]}_{x[0]}\, h[n] + \underbrace{X_e[1]}_{0}\, h[n-1] + \underbrace{X_e[2]}_{0}\, h[n-2] + X_e[3]\, h[n-3]$$

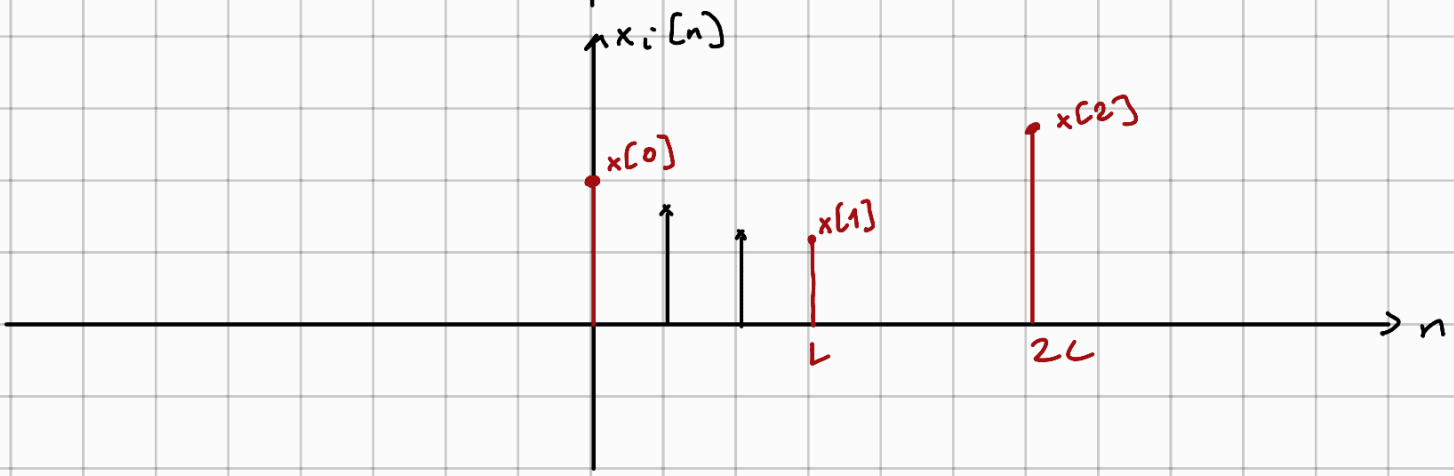
$$+ \underbrace{X_e[L]}_{x[1]}\, h[n-L] + \cdots$$

$$+ \underbrace{X_e[2L]}_{x[2]}\, h[n-2L] + \cdots$$

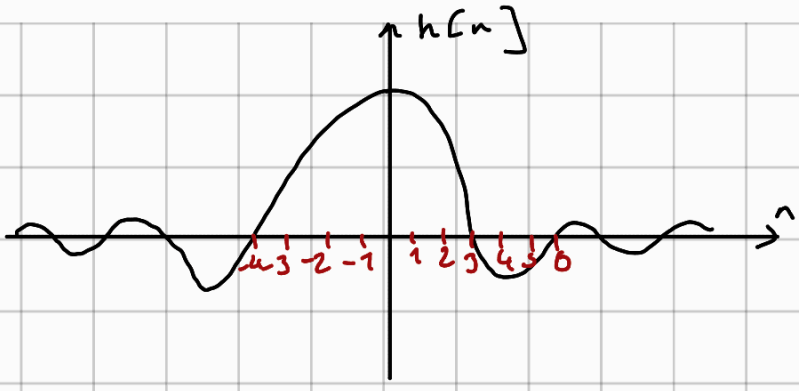
$$\vdots$$

$$x_i[n] = \sum_{k=-\infty}^{\infty} X[k]\, \frac{\sin \pi (n-kL)/L}{\pi (n-kL)/L}$$

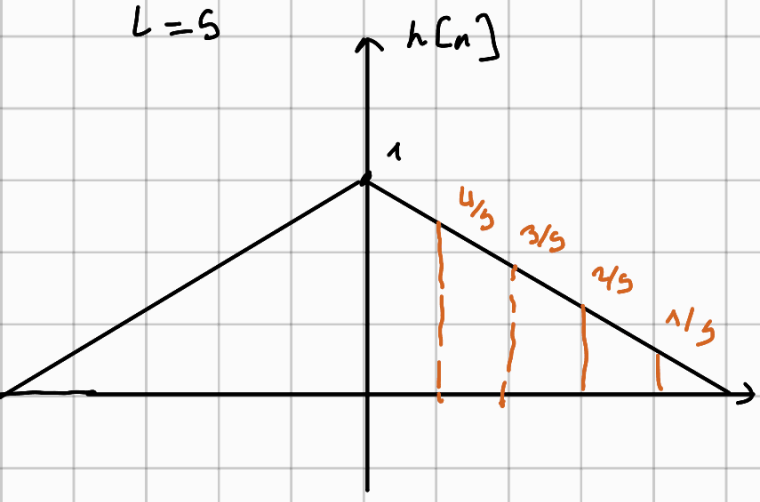


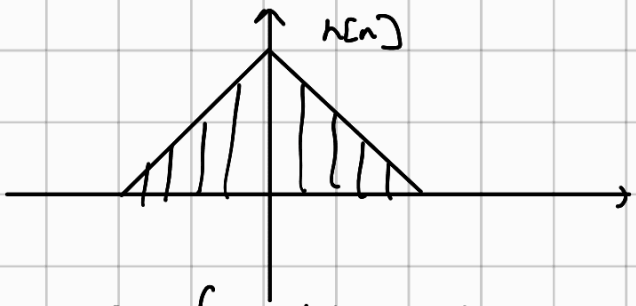


$x_i[n] = x_e[n] * h[n]$



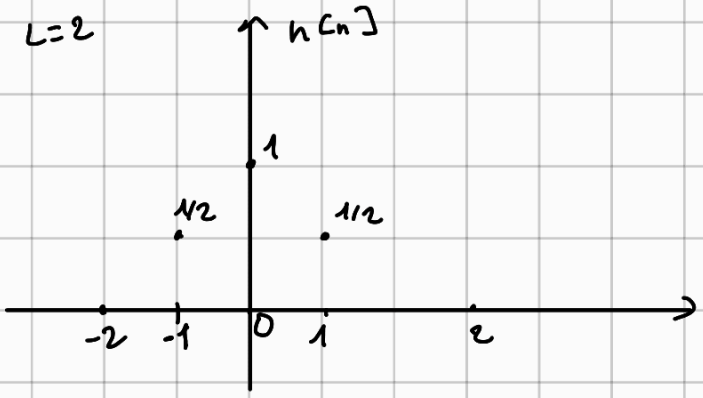
$L = 5$

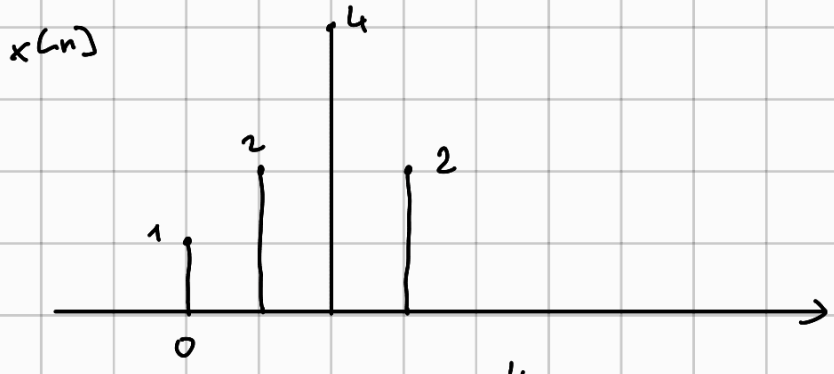




$$h[n] = \begin{cases} 1 - \frac{|n|}{L} & |n| < L \\ 0 & \text{ow} \end{cases}$$

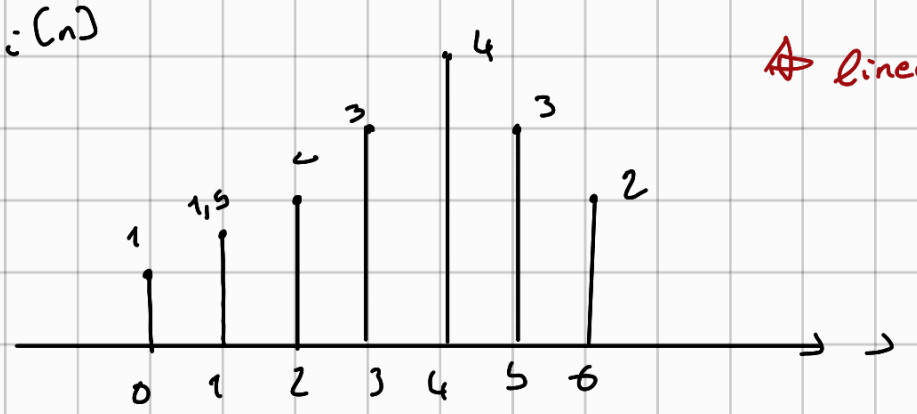
$L = 2$







$x_i[n]$

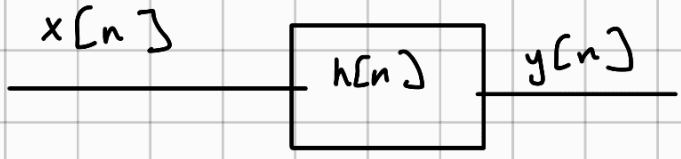


⚠ linear interpolation



Herzaman up gapılır başka

Frequency Response of LTI systems



$$y[n] = x[n] * h[n]$$

$$Y(z) = X(z)\,H(z)$$

$$Y(e^{j\omega}) = X(e^{j\omega})\,H(e^{j\omega})$$
$$|Y(e^{j\omega})| = |X(e^{j\omega})|\,|H(e^{j\omega})|$$
$$\angle Y(e^{j\omega}) = \angle|X(e^{j\omega})| + \angle|H(e^{j\omega})|$$

$$x[n] = A\cos(\omega_0 n + \phi)$$
$$y[n] = A\,|H(e^{j\omega})|\cos(\omega_0 n + \phi + \angle H(e^{j\omega}))$$
$$h_{id}[n] = \delta[n - n_\phi]$$

$$H_{id}(e^{j\omega}) = e^{-j\omega n_d}$$

$$|H_{id}(e^{j\omega})| = 1$$

$$\angle H_{id}(e^{j\omega}) = -\omega_0 n_d$$

$$y[n] = A\cos(\omega_0 n + \phi - \omega_0 n_d)$$

$$= A\cos(\omega_0 (n - n_d) + \phi)$$